OUR PROTOTYPE

A splint made for one astronaut

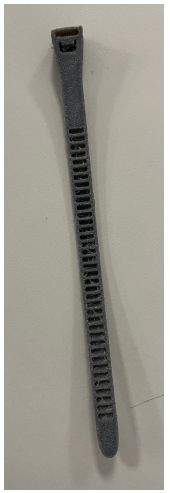
Our value: OrthoPrint gives an injured astronaut a splint made for their own ankle, printed when it's needed and strapped on with one hand.



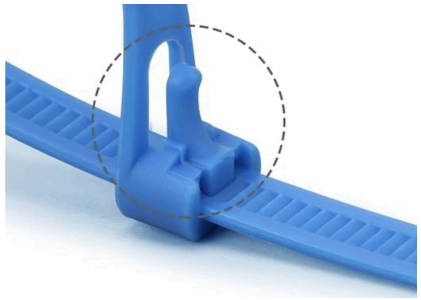
Strap, front



Side



Back

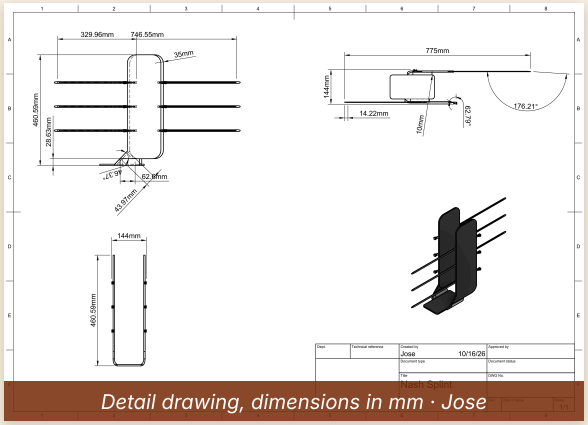


The reusable zip tie behind our closure

See the full CAD
gertrudish978-pixel.github.io/lunar-splint

OTHER DESIGN FEATURES

Details & next steps



Detail drawing, dimensions in mm · Jose

Materials

PLA · ABS · TPU · PETG

CF-PEEK for flight

Size	460.59 mm tall, side plates 144 mm apart
Fit	Unlatches and goes on from above
Shell	Solid TPU wall, firm but moulds slightly

Next steps

- ○ Test strap thickness and break load
- ○ Find where the filament softens with heat
- ○ Build prototype #2 for the CDR

Gladys Porter Early College High School
Brownsville, Texas

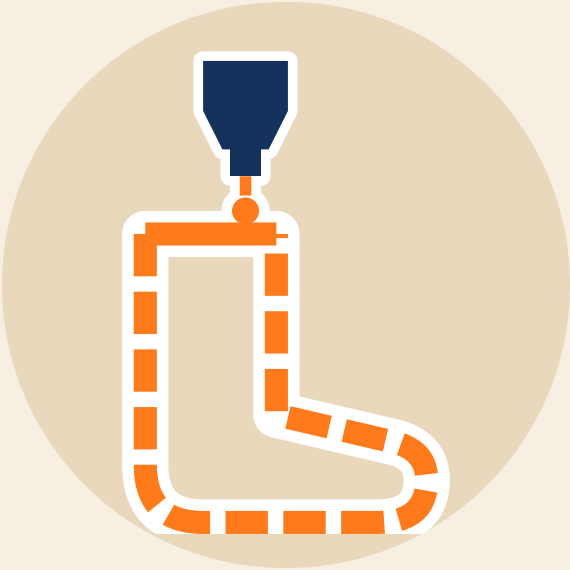
A student design project. Not a medical device.

LUNOVA

NASA HUNCH · LUNAR SPLINT DESIGN CHALLENGE

OrthoPrint

A custom-fit ankle splint made from a scan of the patient's leg.



Team Lunova · Gertrudis, Jose & Nancy
Gladys Porter Early College High School
Brownsville, Texas
Teachers: Lupita C. Vasquez and Claudia Cortez

Design & Prototype · 2026–27

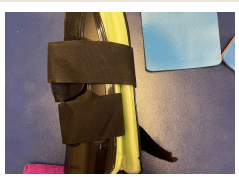
Website, CAD & videos
gertrudish978-pixel.github.io/lunar-splint

PROTOTYPE HIGHLIGHTS

How we got here

1 Research

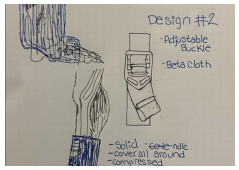
Surveyed 56 students and studied braces with our school's athletic trainer.



Trainer's ankle brace

2 Brainstorming

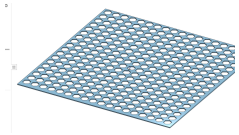
Sketched a one-size cast, then a two-piece splint with a buckle.



Concept sketches

3 Design

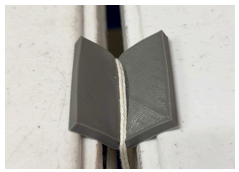
Drew hole patterns and test samples in CAD.



CAD test sample

4 Testing

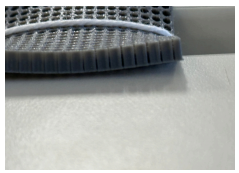
Hung weights on PETG and TPU samples. The holes split the layers.



Hanging-weight test

5 Prototype

Chose a solid TPU wall with zip-tie straps at key points.



Solid TPU wall

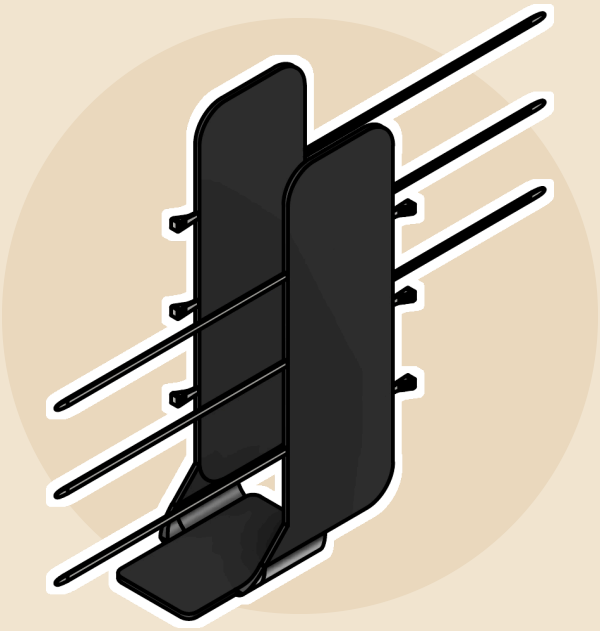
WHAT MAKES IT UNIQUE

Our three innovations

1 Single-piece ratchet closure
Tightens or releases with one hand. No loose parts to lose.

2 Integrated hygiene
Comes off to clean the splint and check the skin.

3 Thermal phase design
Flexible to put on, firm to hold the ankle still.



Our 3D CAD (isometric) · Jose

BEYOND THE REQUIREMENTS

More than the challenge asks

3	innovations, where the challenge asks for one
1	splint per patient, custom-fit from a scan instead of fixed sizes
56	students in our Needs Assessment Survey, plus an athletic-trainer interview
4	print materials tested, with CF-PEEK picked for flight

What our survey told us

Ankle injured most	**63%**
Hard to find a fit	**54%**
On their feet a lot	**96%**
Devices help healing	**98%**

Why it matters

$1,000	to launch every gram, so no box of spare sizes
1	hand is all it takes to tighten or release the straps